

# FACTSHEET FS07 · Size

**What changed on 2 October 2026.** Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS\_RERUN\_2026-10-02.md.

**US, 1999–2012 (pre-registered second period, run 2 October 2026).** The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net: t +1.88 (2013–2026: -0.55), no 2013–2026 effect to test. Long-only alpha vs the equal-weight universe: +1.3% a year, t +0.39 (2013–2026: -2.66), reversed. Registered verdicts are unchanged. Pre-registration, method and every result: PREREG\_US\_1999\_2012.md.

## Do smaller companies beat bigger ones?

Henrik Gjerming · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

|                                                    |                                                                                      |                                                           |
|----------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------|
| STRATEGY                                           | ORIGIN                                                                               | FAMILY                                                    |
| Size premium (Fama-French SMB)                     | Banz (1981)                                                                          | Size                                                      |
| UNIVERSES                                          | BACKTEST                                                                             | BOOKS                                                     |
| US · EU · UK · DK · SCANDI · World (point-in-time) | Feb 2013 – Aug 2026, monthly, net of costs                                           | Long/short and long-only (smallest names in the universe) |
| REBALANCE                                          | STATUS                                                                               |                                                           |
| Monthly                                            | Registered gate $ t  > 2.87$ over 12 cells; every cell in the programme trial ledger |                                                           |

**The claim.** Small companies earn higher returns than large ones. Banz (1981) found it in US stocks 1936–1975; it became Fama & French's SMB factor, but has been weak since the early 1980s unless junk stocks are controlled for (Asness et al. 2018). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -4.3% to +4.3% a year and was positive in 1 of the six; no universe passes the 2.87 gate. The long-only book (smallest names in the universe) had a higher Sharpe than the equal-weight universe in 1 of the six (UK), with alphas of -6.2% to +6.4%, passing the gate in UK. **What goes wrong (§9):** the main drags are a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (appendix C):** Size; its returns sit closest to the low risk cluster of the signal library. **360° view (appendix D):** nearest neighbours Momentum 12-1, Near 52-week high; after those neighbours and the market, nothing is left (alpha  $|t| < 2$  everywhere): the signal is redundant.

Layout revised 2 October 2026 (verdict and scorecard, Sharpe anatomy, fit with the other strategies; detail moved to the appendix). No number changed.

# 1. Verdict and scorecard

**Verdict.** Of the 12 registered cells (two books × six universes), **1 pass** the  $|t| > 2.87$  gate (long-only UK). Programme-wide (772 trials,  $|t| > 3.99$ ): long-only alpha vs EW, UK  $t +4.07$ . US 1999–2012, a period the factsheet was not built on: L/S  $t +1.88$ , long-only alpha  $t +0.39$ .

| Universe     | L/S $t$ | L/S Sharpe | Long-only alpha $t$ | Long-only Sharpe vs equal-weight | Long-only max DD |
|--------------|---------|------------|---------------------|----------------------------------|------------------|
| US           | -0.55   | -0.13      | -2.66               | 0.53 vs 0.85                     | -45%             |
| EU           | -1.19   | -0.37      | -1.61               | 0.41 vs 0.74                     | -29%             |
| UK           | +1.52   | 0.43       | +4.07 ✓             | 1.06 vs 0.67                     | -25%             |
| DK           | -0.23   | -0.06      | -0.23               | 0.71 vs 0.83                     | -38%             |
| SCANDI       | -0.02   | -0.01      | -0.91               | 0.71 vs 0.89                     | -29%             |
| World        | -0.38   | -0.10      | -1.31               | 0.58 vs 0.71                     | -37%             |
| US 1999–2012 | +1.88   | 0.55       | +0.39               | 0.49 vs 0.49                     | –                |

✓ passes the 2.87 gate (Bonferroni over 12 cells); ✗ significantly negative. 2013–2026 unless stated.

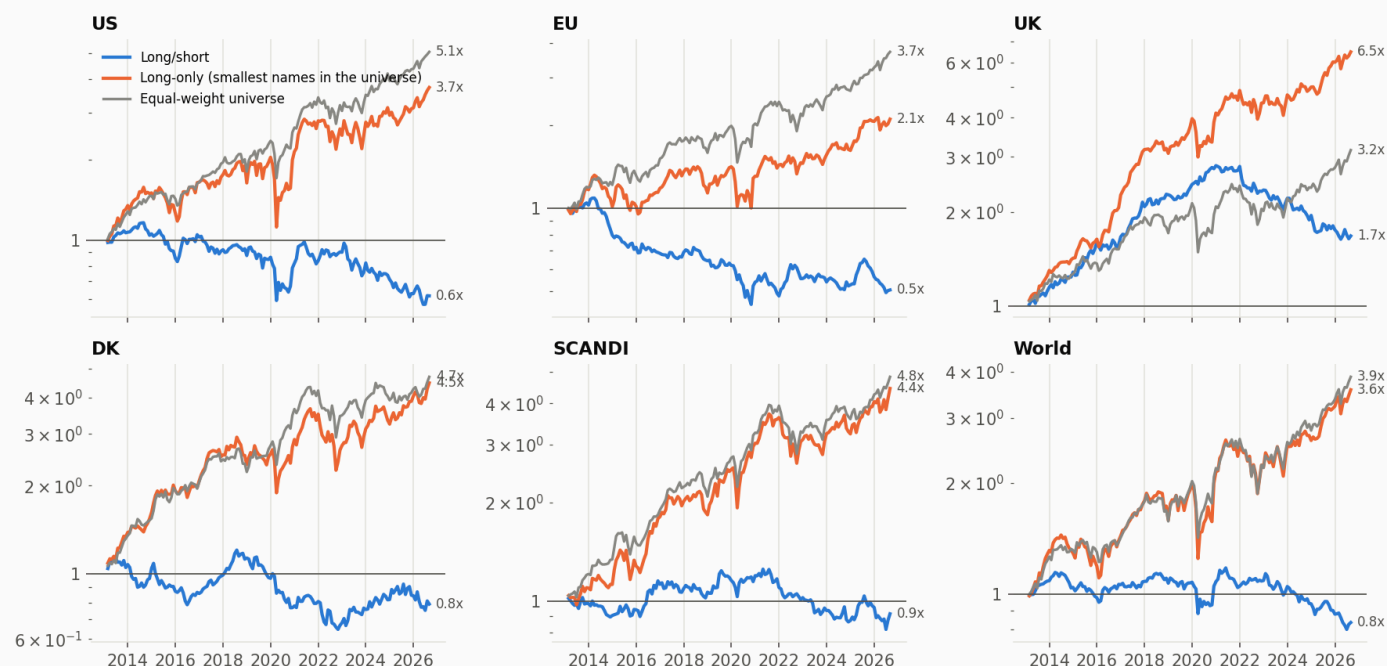
# 2. Headline performance

| Universe | Names / groups | L/S return / yr | L/S $t$ | L/S CAPM alpha ( $t$ ) | L/S beta | Long-only CAGR | Long-only Sharpe | Long-only max DD | EW CAGR | EW Sharpe | Size-weighted CAGR |
|----------|----------------|-----------------|---------|------------------------|----------|----------------|------------------|------------------|---------|-----------|--------------------|
| US       | 499 / 10       | -2.2%           | -0.55   | -10.4% (-2.8)          | 0.62     | +10.2%         | 0.53             | -45%             | +12.7%  | 0.85      | +14.9%             |
| EU       | 260 / 10       | -4.3%           | -1.19   | -6.1% (-1.5)           | 0.16     | +5.7%          | 0.41             | -29%             | +10.1%  | 0.74      | +10.2%             |
| UK       | 330 / 10       | +4.3%           | 1.52    | +3.1% (1.0)            | 0.13     | +14.8%         | 1.06             | -25%             | +8.8%   | 0.67      | +7.8%              |
| DK       | 20 / 3         | -0.8%           | -0.23   | -1.8% (-0.5)           | 0.08     | +11.7%         | 0.71             | -38%             | +12.1%  | 0.83      | +10.2%             |
| SCANDI   | 70 / 5         | -0.1%           | -0.02   | -2.5% (-0.9)           | 0.19     | +11.6%         | 0.71             | -29%             | +12.3%  | 0.89      | +11.0%             |
| World    | 1088 / 10      | -0.9%           | -0.38   | -4.5% (-2.0)           | 0.32     | +9.8%          | 0.58             | -37%             | +10.5%  | 0.71      | +10.3%             |

Monthly, local currency (World in USD). L/S = equal-weighted smallest names in the universe minus largest names, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West  $t$  (6 lags). CAPM vs the equal-weight

universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

#### Size on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjering · Project 10 factsheets

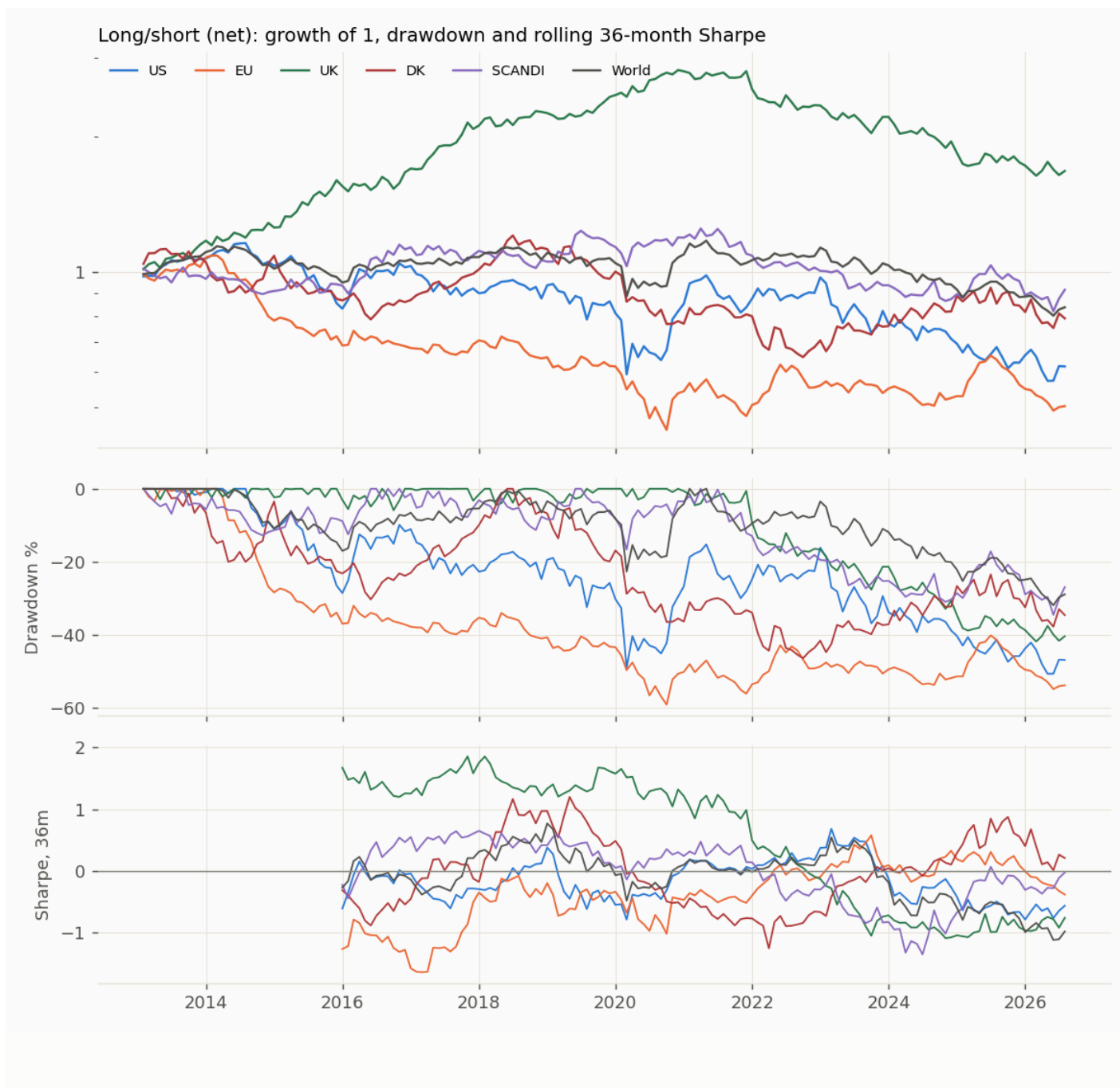
**Read with care.** A long/short book with a negative beta shows a large CAPM alpha in a rising market. The factor-model alpha in section 7, after momentum, value and the other themes, is the better guide: here the L/S CAPM alpha ranges -10.4% to +3.1% and the factor alpha -5.5% to +5.4%.

## 3. Where the Sharpe comes from

Sharpe = annual return ÷ annual volatility. The return splits into the part the market explains (beta × market return) and the rest (alpha, net of costs); the volatility into the market's share and the strategy's own. The last column is the Sharpe the book would have with its market exposure hedged out (before hedging costs).

| Universe | Book      | Return / yr | = market part | + alpha | Costs / yr (inside the return) | Volatility | of which market | Sharpe | Sharpe, market hedged |
|----------|-----------|-------------|---------------|---------|--------------------------------|------------|-----------------|--------|-----------------------|
| US       | L/S       | -2.2%       | +8.2%         | -10.4%  | -0.6%                          | 16.4%      | 9.7%            | -0.13  | -0.79                 |
| US       | Long-only | +12.6%      | +18.7%        | -6.1%   | —                              | 23.9%      | 22.1%           | 0.53   | -0.66                 |
| EU       | L/S       | -4.3%       | +1.8%         | -6.1%   | -0.5%                          | 11.8%      | 2.3%            | -0.37  | -0.53                 |
| EU       | Long-only | +6.9%       | +11.1%        | -4.2%   | —                              | 16.9%      | 14.9%           | 0.41   | -0.52                 |
| UK       | L/S       | +4.3%       | +1.2%         | +3.1%   | -0.7%                          | 9.9%       | 1.8%            | 0.43   | 0.32                  |
| UK       | Long-only | +14.8%      | +8.5%         | +6.3%   | —                              | 13.9%      | 12.6%           | 1.06   | 1.06                  |





## 4. How it fits next to the market and the other strategies

A long/short book improves a market portfolio when its own Sharpe beats the hurdle: its correlation with the market  $\times$  the market's Sharpe. A negative correlation makes the hurdle negative, so even a book with a small positive Sharpe diversifies; the size of the gain still depends on that Sharpe.

| Universe | Corr. with market | Avg corr. with the other strategies | Most similar strategy | Market Sharpe | Hurdle Sharpe | This L/S Sharpe | Verdict | L/S in the worst 10% of market months | Market in those months |
|----------|-------------------|-------------------------------------|-----------------------|---------------|---------------|-----------------|---------|---------------------------------------|------------------------|
| US       | +0.59             | -0.29                               | 52-week high (-0.79)  | 0.85          | +0.50         | -0.13           | dilutes | -3.2%                                 | -7.3%                  |

| Universe | Corr. with market | Avg corr. with the other strategies | Most similar strategy          | Market Sharpe | Hurdle Sharpe | This L/S Sharpe | Verdict | L/S in the worst 10% of market months | Market in those months |
|----------|-------------------|-------------------------------------|--------------------------------|---------------|---------------|-----------------|---------|---------------------------------------|------------------------|
| EU       | +0.20             | -0.17                               | Low volatility (-0.37)         | 0.75          | +0.15         | -0.37           | dilutes | -0.7%                                 | -6.5%                  |
| UK       | +0.18             | +0.03                               | Same-month seasonality (+0.28) | 0.68          | +0.12         | 0.43            | adds    | -1.6%                                 | -6.6%                  |
| DK       | +0.09             | +0.02                               | Buy at the 52-week low (+0.30) | 0.83          | +0.07         | -0.06           | dilutes | -1.2%                                 | -7.2%                  |
| SCANDI   | +0.25             | -0.05                               | Low volatility (-0.32)         | 0.89          | +0.22         | -0.01           | dilutes | -1.7%                                 | -6.9%                  |
| World    | +0.56             | -0.29                               | 52-week high (-0.70)           | 0.71          | +0.39         | -0.10           | dilutes | -1.9%                                 | -7.5%                  |

*Long/short (net), monthly, 2013–2026; market = equal-weight universe; other strategies = the long/short books of FS01–FS12. The Strategy Cockpit lets you build books of several strategies.*

## 5. Strategy description

Small companies earn higher returns than large ones. Banz (1981) found it in US stocks 1936–1975; it became Fama & French's SMB factor, but has been weak since the early 1980s unless junk stocks are controlled for (Asness et al. 2018).

**Why it might work.** Risk (distress, illiquidity; Fama & French 1993) or mispricing; much of the historical premium came from the smallest, least liquid stocks and January.

| Rule      | Original (Banz (1981))    | This factsheet                                                                              |
|-----------|---------------------------|---------------------------------------------------------------------------------------------|
| Signal    | Market capitalisation     | US: market cap (Sharadar). Elsewhere: 63-day average traded value (price × volume), a proxy |
| Portfolio | Size deciles or SMB (2×3) | Extreme quantile L/S within the universe, monthly                                           |
| Weighting | Value weighted            | Equal weighted                                                                              |

## 6. Method: data, signal and portfolio

### Data load

| Universe | Prices                                                                  | Membership                                                                                                                                                                                                                | Names eligible (median) | Currency                   |
|----------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|
| US       | Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted       | <b>Point-in-time:</b> the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)                                                                                                           | ~503                    | USD                        |
| EU       | Project1 yfinance cache, Adj Close (TR)                                 | <b>Point-in-time:</b> union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced | ~238                    | local (EUR, SEK, DKK, PLN) |
| UK       | Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×) | <b>Point-in-time:</b> FTSE 100 + FTSE 250, quarterly snapshots; 75% priced                                                                                                                                                | ~242                    | GBP                        |
| DK       | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMX Copenhagen 25, quarterly snapshots; 91% priced                                                                                                                                                  | ~19                     | DKK                        |
| SCANDI   | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced                                                                                                                                 | ~62                     | local (DKK, SEK, EUR)      |
| World    | US Sharadar + UK + EU above                                             | <b>Point-in-time</b> union of the three; local-currency returns summed, not converted                                                                                                                                     | ~963                    | mixed                      |

*Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.*

- **Data gap:** No shares outstanding outside the US: size is proxied by traded value, which also measures liquidity and turnover.
- **Data gap:** The universes are blue chips; the classic size premium lives in micro and small caps, which are excluded by construction.

### Signal creation

$SIZE_{i,t}$  = log market cap (US) or log 63-day average traded value (elsewhere) at month-end  $t$ ; World ranks within each region first. Signals use only information up to month-end  $t$  and are traded in month  $t+1$ . Eligibility: in the point-in-time universe on any of the last five trading days of month  $t$ .

### Model build

- Sort eligible stocks into equal-count groups (deciles  $\geq 100$  names, quintiles 50–99, terciles  $< 50$ ; fixed per universe).

- **Long/short** = smallest names in the universe minus largest names; **long-only** = smallest names in the universe; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

## 7. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

### Long/short

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                                                                                   |
|----------|------------------------|------------|----------|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | -2.6%      | -1.12    | 0.78           | mkt +0.22 (+3.8), size +0.44 (+3.6), value +0.75 (+6.7), momentum -0.22 (-2.5), low_risk -0.37 (-2.3), quality -0.62 (-4.1) |
| EU       | French Europe 5F + WML | -5.2%      | -1.60    | 0.20           | SMB +0.47 (+3.4), HML +0.36 (+2.1)                                                                                          |
| UK       | JKP GBR 7 themes       | +5.4%      | 2.29     | 0.41           | size +0.97 (+8.2), value -0.46 (-3.5)                                                                                       |
| DK       | JKP DNK 7 themes       | -1.9%      | -0.52    | 0.15           | size +0.53 (+3.0), low_risk +0.32 (+2.3)                                                                                    |
| SCANDI   | French Europe 5F + WML | -0.7%      | -0.30    | 0.10           | SMB +0.47 (+2.4)                                                                                                            |
| World    | JKP World 7 themes     | -5.5%      | -2.71    | 0.60           | mkt +0.12 (+3.4), size +0.63 (+5.2), value +0.42 (+4.0), short_term_reversal +0.35 (+2.4)                                   |

### Long-only

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                                                                                                   |
|----------|------------------------|------------|----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | +2.1%      | 1.16     | 0.94           | mkt +1.07 (+23.7), value +0.69 (+7.1), momentum -0.25 (-4.9), low_risk -0.33 (-2.3), quality -0.40 (-2.7), short_term_reversal +0.41 (+2.2) |
| EU       | French Europe 5F + WML | +0.9%      | 0.37     | 0.64           | Mkt-RF +0.74 (+10.4), HML +0.43 (+2.5)                                                                                                      |
| UK       | JKP GBR 7 themes       | +11.1%     | 4.38     | 0.70           | mkt +0.51 (+8.8), size +0.31 (+2.9), low_risk -0.36 (-2.6)                                                                                  |
| DK       | JKP DNK 7 themes       | +4.1%      | 1.05     | 0.61           | mkt +0.77 (+9.8), momentum -0.20 (-2.2)                                                                                                     |
| SCANDI   | French Europe 5F + WML | +7.1%      | 2.30     | 0.51           | Mkt-RF +0.76 (+8.9)                                                                                                                         |

| Universe | Factor model       | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                                       |
|----------|--------------------|------------|----------|----------------|---------------------------------------------------------------------------------|
| World    | JKP World 7 themes | +2.8%      | 1.15     | 0.87           | mkt +0.77 (+14.3), size +0.33 (+3.3), value +0.51 (+4.4), low_risk -0.23 (-2.2) |

## 8. Statistical detail

- **Gate:**  $|t| > 2.87$  (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 1 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -10.4% to +3.1%,  $|t|$  past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 1 of the six; alpha -6.2% to +6.4%, passing in 1 of 6.
- **Probabilistic Sharpe** of the L/S,  $P(\text{true Sharpe} > 0)$ : US 0.31, EU 0.10, UK 0.94, DK 0.42, SCANDI 0.49, World 0.36.
- **Publication decay:** gross L/S -5.9% to +14.1% a year in 2013–19 against -4.4% to -1.2% in 2020–26.

## 9. What goes wrong, and how to fix it

**Method.** Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FSO3–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FSO3–FS12, gate  $p < 0.0020$ .

### 9.1 Diagnosis (design window 2013–2019)

| Universe | L/S gross / yr | Beta  | Beta drag / yr | CAPM alpha (gross) | Costs / yr | Long leg vs EW / yr | EW vs short leg / yr | Turnover / month | Worst months                           |
|----------|----------------|-------|----------------|--------------------|------------|---------------------|----------------------|------------------|----------------------------------------|
| US       | -0.9%          | 0.53  | +6.9%          | -7.8%              | -0.3%      | -0.9%               | +0.1%                | 14%              | 2019-08 -10%, 2015-07 -6%, 2015-12 -6% |
| EU       | -5.9%          | 0.07  | +0.7%          | -6.6%              | -0.3%      | -4.5%               | -1.4%                | 13%              | 2014-05 -6%, 2014-10 -6%, 2015-11 -5%  |
| UK       | +14.1%         | -0.19 | -2.2%          | +16.4%             | -0.4%      | +9.4%               | +4.7%                | 19%              | 2018-04 -4%, 2016-10 -4%, 2017-12 -3%  |
| DK       | +0.8%          | -0.08 | -1.2%          | +2.0%              | -0.3%      | -0.6%               | +1.4%                | 15%              | 2015-02 -8%, 2014-02 -7%, 2014-05 -7%  |
| SCANDI   | +3.1%          | 0.17  | +2.7%          | +0.4%              | -0.2%      | -0.6%               | +3.6%                | 9%               | 2015-09 -5%, 2013-10 -4%, 2017-07 -4%  |

| Universe | L/S gross / yr | Beta | Beta drag / yr | CAPM alpha (gross) | Costs / yr | Long leg vs EW / yr | EW vs short leg / yr | Turnover / month | Worst months                          |
|----------|----------------|------|----------------|--------------------|------------|---------------------|----------------------|------------------|---------------------------------------|
| World    | +1.8%          | 0.23 | +2.5%          | -0.6%              | -0.3%      | +0.4%               | +1.5%                | 14%              | 2019-08 -5%, 2014-11 -3%, 2018-12 -3% |

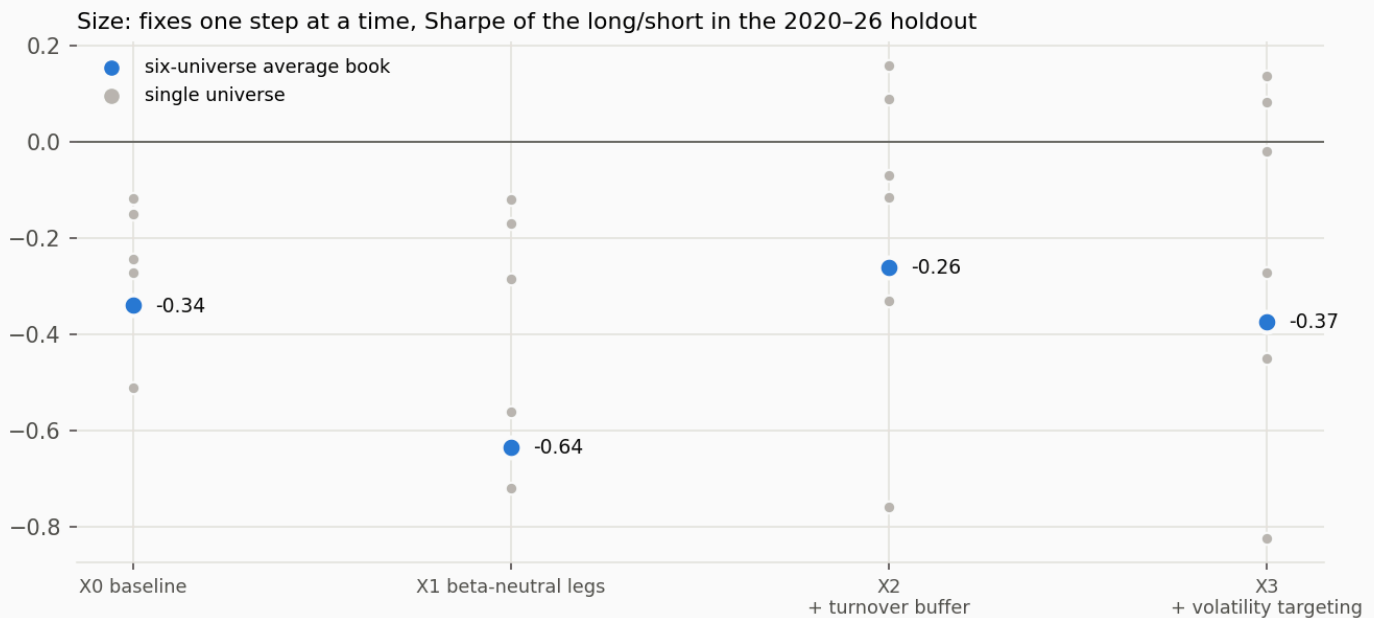
On average across the six universes the gross spread was +2.2% a year, of which the market exposure (beta +0.12) contributed +1.6%; the beta-adjusted spread (CAPM alpha) was +0.6%. Costs took 0.3% a year at 14% monthly turnover across both legs. The long leg beat the universe by +0.5% and the short leg lagged it by +1.6% a year, so most of the spread comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +0.3%, +0.2%, -0.9% a year.

## 9.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

| Step                      | Design Sharpe (avg) | Δ design (p)    | Holdout Sharpe (book) | Δ holdout (p)   | Universes improved | Holdout return / yr | Holdout max DD | Holdout alpha vs EW (t) | Costs / yr |
|---------------------------|---------------------|-----------------|-----------------------|-----------------|--------------------|---------------------|----------------|-------------------------|------------|
| X0 baseline               | 0.19                | –               | -0.34                 | –               | –                  | -3.0%               | -27%           | -6.1% (-2.0)            | 0.3%       |
| X1 beta-neutral legs      | 0.25                | +0.26 (p 0.035) | -0.64                 | -0.30 (p 0.994) | 1 of 6             | -5.3%               | -34%           | -7.3% (-2.3)            | 0.4%       |
| X2 + turnover buffer      | 0.37                | +0.23 (p 0.081) | -0.26                 | +0.37 (p 0.020) | 5 of 6             | -1.6%               | -18%           | -2.7% (-1.2)            | 0.1%       |
| X3 + volatility targeting | 0.34                | -0.25 (p 0.753) | -0.37                 | -0.11 (p 0.882) | 1 of 6             | -3.0%               | -22%           | -4.7% (-1.5)            | 0.1%       |

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



- **X1 beta-neutral legs:** +0.26 design, -0.30 holdout; helps in one window and hurts in the other: not reliable.
- **X2 + turnover buffer:** +0.23 design, +0.37 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.25 design, -0.11 holdout; hurts in both windows.

**Where it ends:** holdout Sharpe -0.34 → -0.37, return -3.0% → -3.0% a year (six-universe book).

## 10. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** No shares outstanding outside the US: size is proxied by traded value, which also measures liquidity and turnover.
5. **Data gap.** The universes are blue chips; the classic size premium lives in micro and small caps, which are excluded by construction.
6. **Trials.** Every gated cell is in the factsheet trial ledger; the fix ladder is counted inside FS03–FS12 (25 trials).

## 11. Academic references

- Banz, R. (1981). The relationship between return and market value of common stocks. *Journal of Financial Economics*, 9(1), 3–18.

- Fama, E. & French, K. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- van Dijk, M. (2011). Is size dead? A review of the size effect in equity returns. *Journal of Banking & Finance*, 35(12), 3263–3274.
- Asness, C., Frazzini, A., Israel, R., Moskowitz, T. & Pedersen, L. H. (2018). Size matters, if you control your junk. *Journal of Financial Economics*, 129(3), 479–509.
- Newey, W. & West, K. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Politis, D. & Romano, J. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Bailey, D. & López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- Jensen, T. I., Kelly, B. & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- Fama, E. & French, K. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- McLean, R. D. & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R. & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Barroso, P. & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Treynor, J. & Mazuy, K. (1966). Can mutual funds outguess the market? *Harvard Business Review*, 44(4), 131–136.

## 12. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse\_xs.py FS07 → build\_xs.py FS07.  
Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.

## Appendix

### A. Performance in detail

**US (S&P 500 members, point-in-time)**, median 499 names, 10 groups

| Book      | Mean / yr | CAGR  | Vol   | Sharpe [95% CI]        | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t)     | CVaR 95% (month) |
|-----------|-----------|-------|-------|------------------------|---------|--------|-----|---------|------|------------------|------------------|
| L/S (net) | -2.2%     | -3.5% | 16.4% | -0.13<br>[-0.58, 0.33] | -0.30   | -51%   | 47% | -0.55   | 0.62 | -10.4%<br>(-2.8) | 9.5%             |

| Book                                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]   | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t) | CVaR 95% (month) |
|----------------------------------------|-----------|--------|-------|-------------------|---------|--------|-----|---------|------|--------------|------------------|
| Long-only (net)                        | +12.6%    | +10.2% | 23.9% | 0.53 [0.10, 1.05] | 0.67    | -45%   | 61% | 2.41    | 1.42 | -6.2% (-2.7) | 13.6%            |
| Smallest names in the universe (gross) | +12.8%    | +10.4% | 23.9% | 0.54 [0.11, 1.06] | 0.68    | -45%   | 61% | 2.45    | 1.42 | -6.0% (-2.6) | 13.6%            |
| Largest names (gross)                  | +14.5%    | +14.4% | 13.7% | 1.05 [0.60, 1.60] | 1.75    | -26%   | 68% | 5.33    | 0.80 | +3.9% (2.2)  | 7.7%             |
| EW universe                            | +13.2%    | +12.7% | 15.5% | 0.85 [0.42, 1.42] | 1.30    | -27%   | 68% | 4.26    | 1.00 | –            | 9.2%             |
| Size-weighted universe                 | +14.9%    | +14.9% | 14.2% | 1.05 [0.60, 1.61] | 1.73    | -24%   | 69% | 5.29    | 0.86 | +3.6% (2.3)  | 8.3%             |

EU (11 national blue-chip indices, point-in-time), median 260 names, 10 groups

| Book                                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]     | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t) | CVaR 95% (month) |
|----------------------------------------|-----------|--------|-------|---------------------|---------|--------|-----|---------|------|--------------|------------------|
| L/S (net)                              | -4.3%     | -4.9%  | 11.8% | -0.37 [-1.00, 0.17] | -0.60   | -59%   | 39% | -1.19   | 0.16 | -6.1% (-1.5) | 6.2%             |
| Long-only (net)                        | +6.9%     | +5.7%  | 16.9% | 0.41 [-0.04, 0.92]  | 0.52    | -29%   | 57% | 1.66    | 1.03 | -4.1% (-1.6) | 10.2%            |
| Smallest names in the universe (gross) | +7.1%     | +5.8%  | 16.9% | 0.42 [-0.03, 0.93]  | 0.54    | -29%   | 57% | 1.70    | 1.03 | -4.0% (-1.6) | 10.2%            |
| Largest names (gross)                  | +10.9%    | +10.4% | 13.9% | 0.78 [0.30, 1.34]   | 1.18    | -28%   | 61% | 3.22    | 0.87 | +1.6% (0.8)  | 8.3%             |
| EW universe                            | +10.7%    | +10.1% | 14.5% | 0.74 [0.28, 1.29]   | 1.13    | -26%   | 62% | 3.18    | 1.00 | –            | 8.5%             |
| Size-weighted universe                 | +10.6%    | +10.2% | 13.7% | 0.77 [0.32, 1.29]   | 1.21    | -25%   | 58% | 3.37    | 0.91 | +0.9% (0.7)  | 7.9%             |

UK (FTSE 350, point-in-time), median 330 names, 10 groups

| Book                                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]    | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t) | CVaR 95% (month) |
|----------------------------------------|-----------|--------|-------|--------------------|---------|--------|-----|---------|------|--------------|------------------|
| L/S (net)                              | +4.3%     | +3.9%  | 9.9%  | 0.43 [-0.18, 1.07] | 0.62    | -42%   | 55% | 1.52    | 0.13 | +3.1% (1.0)  | 5.0%             |
| Long-only (net)                        | +14.8%    | +14.8% | 13.9% | 1.06 [0.55, 1.74]  | 1.78    | -25%   | 67% | 4.29    | 0.89 | +6.4% (4.1)  | 8.2%             |
| Smallest names in the universe (gross) | +15.2%    | +15.2% | 13.9% | 1.09 [0.57, 1.77]  | 1.83    | -24%   | 68% | 4.39    | 0.89 | +6.7% (4.3)  | 8.2%             |
| Largest names (gross)                  | +10.1%    | +9.7%  | 12.9% | 0.79 [0.24, 1.44]  | 1.18    | -26%   | 64% | 2.97    | 0.76 | +2.9% (1.2)  | 7.6%             |



| Book                                         | Mean /<br>yr | CAGR   | Vol   | Sharpe<br>[95% CI]        | Sortino | Max<br>DD | Hit | t(mean) | Beta | Alpha/yr<br>(t) | CVaR 95%<br>(month) |
|----------------------------------------------|--------------|--------|-------|---------------------------|---------|-----------|-----|---------|------|-----------------|---------------------|
| L/S (net)                                    | -0.9%        | -1.3%  | 9.3%  | -0.10<br>[-0.60,<br>0.39] | -0.20   | -32%      | 50% | -0.38   | 0.32 | -4.5%<br>(-2.0) | 5.3%                |
| Long-only (net)                              | +11.3%       | +9.8%  | 19.6% | 0.58 [0.12,<br>1.13]      | 0.79    | -37%      | 59% | 2.46    | 1.17 | -1.9%<br>(-1.3) | 11.6%               |
| Smallest names<br>in the universe<br>(gross) | +11.6%       | +10.1% | 19.6% | 0.59 [0.14,<br>1.14]      | 0.81    | -37%      | 59% | 2.51    | 1.17 | -1.7%<br>(-1.1) | 11.6%               |
| Largest names<br>(gross)                     | +11.9%       | +11.4% | 14.1% | 0.84 [0.37,<br>1.43]      | 1.30    | -28%      | 66% | 3.65    | 0.85 | +2.3%<br>(2.0)  | 8.4%                |
| EW universe                                  | +11.3%       | +10.5% | 16.0% | 0.71 [0.25,<br>1.27]      | 1.02    | -30%      | 62% | 3.07    | 1.00 | –               | 9.7%                |
| Size-weighted<br>universe                    | +11.0%       | +10.3% | 14.8% | 0.74 [0.28,<br>1.31]      | 1.10    | -28%      | 61% | 3.19    | 0.91 | +0.7%<br>(0.8)  | 8.9%                |

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | +7.0%  | +2.7%  | +15.2% | +7.8%  | -1.8%  | +8.5%  |
| 2014 | -1.4%  | -22.3% | +11.8% | -2.9%  | -8.0%  | -4.0%  |
| 2015 | -19.0% | -9.6%  | +24.4% | -16.6% | +4.1%  | -6.2%  |
| 2016 | +20.1% | -5.0%  | +4.1%  | -1.7%  | +19.6% | +8.4%  |
| 2017 | -10.3% | +1.0%  | +24.8% | +15.6% | -2.9%  | +2.1%  |
| 2018 | -4.9%  | -7.0%  | +9.1%  | +14.3% | -2.6%  | -1.2%  |
| 2019 | -1.4%  | -3.8%  | +8.0%  | -14.7% | +8.5%  | +0.6%  |
| 2020 | -5.1%  | -13.1% | +14.7% | -19.5% | +3.5%  | +0.3%  |
| 2021 | +2.1%  | -11.0% | -0.6%  | +3.6%  | -8.0%  | -2.8%  |
| 2022 | +7.6%  | +15.8% | -15.9% | -15.4% | -6.8%  | +3.6%  |
| 2023 | -8.8%  | -2.8%  | -5.9%  | +11.0% | -8.7%  | -3.7%  |
| 2024 | -13.3% | -2.5%  | -11.8% | +8.1%  | -6.8%  | -10.2% |
| 2025 | -11.5% | +7.0%  | -10.1% | +3.0%  | +9.6%  | -6.3%  |
| 2026 | -1.9%  | -10.4% | -4.3%  | -6.3%  | -4.4%  | -5.1%  |

Calendar-year returns, long-only (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | +34.6% | +16.8% | +30.4% | +38.8% | +9.5%  | +31.1% |
| 2014 | +11.5% | -13.2% | +9.4%  | +16.8% | +2.6%  | -1.8%  |

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2015 | -16.0% | +0.4%  | +15.0% | +18.7% | +13.8% | -7.7%  |
| 2016 | +30.7% | +16.3% | +38.4% | +8.1%  | +41.4% | +20.8% |
| 2017 | +6.7%  | +14.9% | +39.6% | +22.5% | +11.5% | +24.5% |
| 2018 | -7.0%  | -16.3% | -2.8%  | -7.4%  | -9.1%  | -11.5% |
| 2019 | +24.8% | +24.7% | +29.0% | +8.8%  | +37.5% | +26.0% |
| 2020 | +5.0%  | -7.6%  | +4.2%  | +12.1% | +20.2% | +7.4%  |
| 2021 | +30.4% | +9.8%  | +18.0% | +22.0% | +19.7% | +16.7% |
| 2022 | -11.7% | +1.4%  | -8.7%  | -21.9% | -14.4% | -11.9% |
| 2023 | +12.0% | +7.5%  | +5.9%  | +14.1% | +4.4%  | +14.7% |
| 2024 | +6.1%  | +3.7%  | +6.0%  | +3.7%  | -0.2%  | +3.1%  |
| 2025 | +6.6%  | +27.6% | +20.0% | +24.8% | +25.3% | +23.8% |
| 2026 | +19.2% | +1.9%  | +7.9%  | +10.8% | +9.7%  | +10.8% |

## B. Trading record and current book

The rebalance log per universe is in `results/xs_FS07_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

| Universe | Rebalances | Names long / short | Turnover long leg / month | Turnover short leg / month | L/S months positive | Worst L/S month | When    |
|----------|------------|--------------------|---------------------------|----------------------------|---------------------|-----------------|---------|
| US       | 163        | 50 / 49            | 9%                        | 4%                         | 47%                 | -24.2%          | 2020-03 |
| EU       | 163        | 26 / 26            | 6%                        | 6%                         | 39%                 | -9.4%           | 2020-07 |
| UK       | 163        | 33 / 33            | 15%                       | 5%                         | 55%                 | -8.7%           | 2022-01 |
| DK       | 163        | 7 / 6              | 10%                       | 6%                         | 53%                 | -12.6%          | 2020-03 |
| SCANDI   | 163        | 14 / 14            | 4%                        | 6%                         | 47%                 | -8.6%           | 2024-10 |
| World    | 163        | 109 / 108          | 10%                       | 5%                         | 50%                 | -14.2%          | 2020-03 |

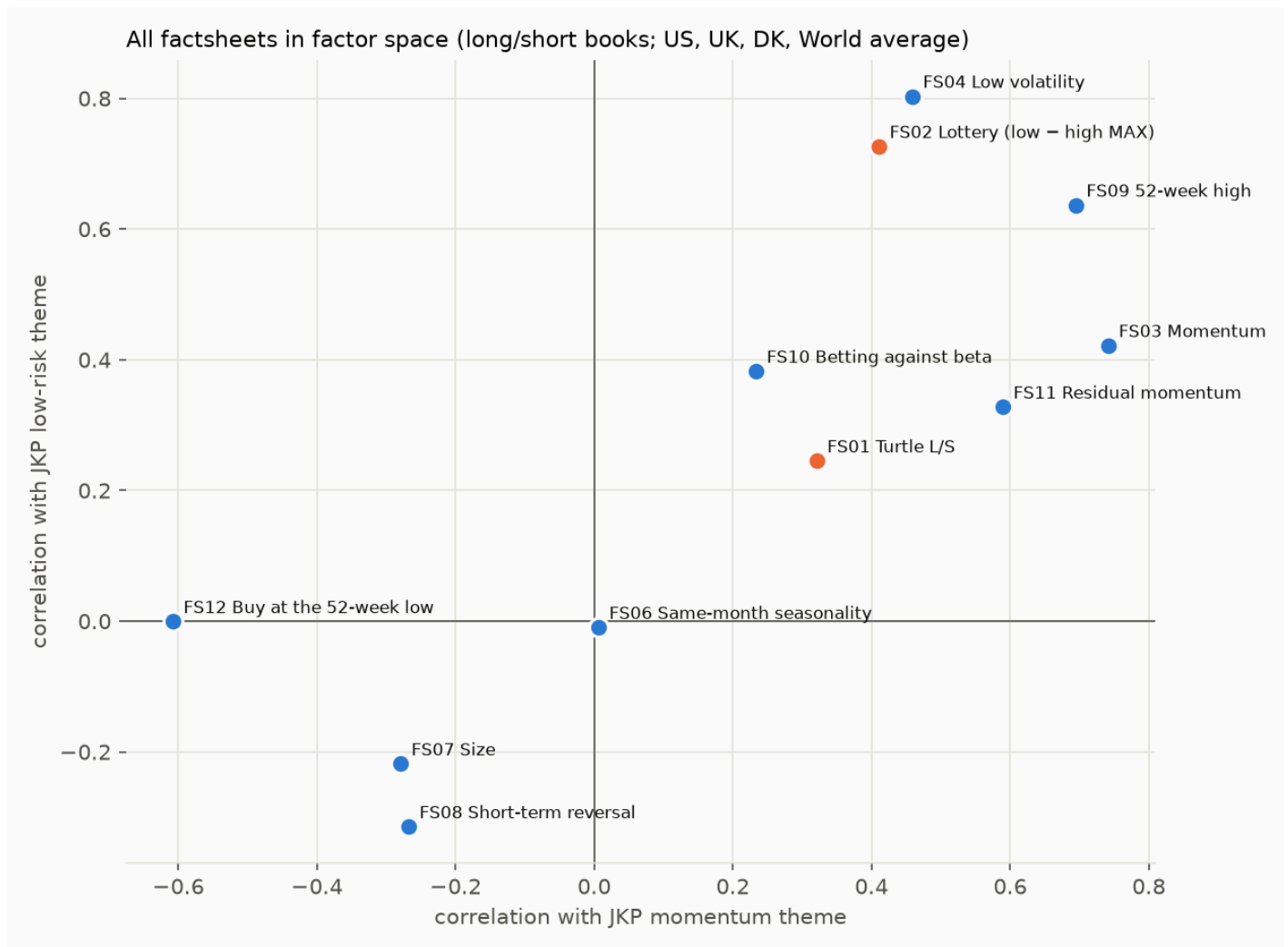
**Current book** (signal at 31 August 2026; first 8 names per leg)

| Universe | Long: smallest names in the universe                                  | Short: largest names                                                              |
|----------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| US       | TTD, TAP, BLDR, NCLH, MOS, CAG, AOS, ARE                              | NVDA, AAPL, GOOGL, MSFT, AMZN, AVGO, META, TSLA                                   |
| EU       | NYR.BR, OBEL.BR, AGFB.BR, GTC.WA, KER.WA, TESB.BR, BPOST.BR, ONTEX.BR | NOVO-B.CO, ASML.AS, VOLV-B.ST, INVE-B.ST, ERIC-B.ST, SAAB-B.ST, AZN.ST, ATCO-A.ST |
| UK       | RTW.L, BPCR.L, PEY.L, MNTN.L, MTLN.L, HANA.L, PRN.L, NAS.L            | AZN.L, HSBA.L, RR.L, SHEL.L, GLEN.L, AAL.L, RIO.L, BP.L                           |

| Universe | Long: smallest names in the universe                                            | Short: largest names                                                             |
|----------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| DK       | NDA-DK.CO, BAVA.CO, GN.CO, RBREW.CO, AMBU-B.CO, ISS.CO, MAERSK-A.CO, JYSK.CO    | NOVO-B.CO, DSV.CO, VWS.CO, MAERSK-B.CO, DANSKE.CO, NSIS-B.CO, GMAB.CO, PNDORA.CO |
| SCANDI   | KALMAR.HE, QTCOM.HE, KEMIRA.HE, HIAB.HE, TYRES.HE, TIETO.HE, MANTA.HE, HUH1V.HE | NOVO-B.CO, VOLV-B.ST, INVE-B.ST, ERIC-B.ST, SAAB-B.ST, AZN.ST, ATCO-A.ST, DSV.CO |

## C. Classification: where does it fit?

| Dimension                         | Size                                                                                                                                                    |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Series family                     | <b>Size</b>                                                                                                                                             |
| Academic style                    | Size premium (Fama-French SMB)                                                                                                                          |
| Signal                            | Price only (daily total returns)                                                                                                                        |
| Cross-section vs time series      | <b>Cross-sectional:</b> stocks ranked against each other each month                                                                                     |
| Horizon and turnover              | One month; 14% of the two legs replaced monthly                                                                                                         |
| Market exposure                   | L/S beta 0.08 to 0.62                                                                                                                                   |
| Payoff shape                      | Treynor–Mazuy $\gamma$ -2.21 to 1.24 (t -2.6 to 0.8); worst 10% of market months -3.2% to -0.7% a month, best 10% -0.9% to +5.2%                        |
| Nearest factor theme              | size (3 of 6 universes), quality (1 of 6 universes)                                                                                                     |
| Nearest library signals (returns) | Momentum 12-1 (3 universes), Volatility (3 universes), Near 52-week high (2 universes)                                                                  |
| Economic rationale                | Risk (distress, illiquidity; Fama & French 1993) or mispricing; much of the historical premium came from the smallest, least liquid stocks and January. |
| Publication                       | Banz 1981; Fama & French 1992–93; 'Is size dead?' (van Dijk 2011); 'Size matters, if you control your junk' (Asness et al. 2018)                        |
| Where it fits                     | In the <b>low risk</b> cluster of the library; fully explained by its neighbours.                                                                       |

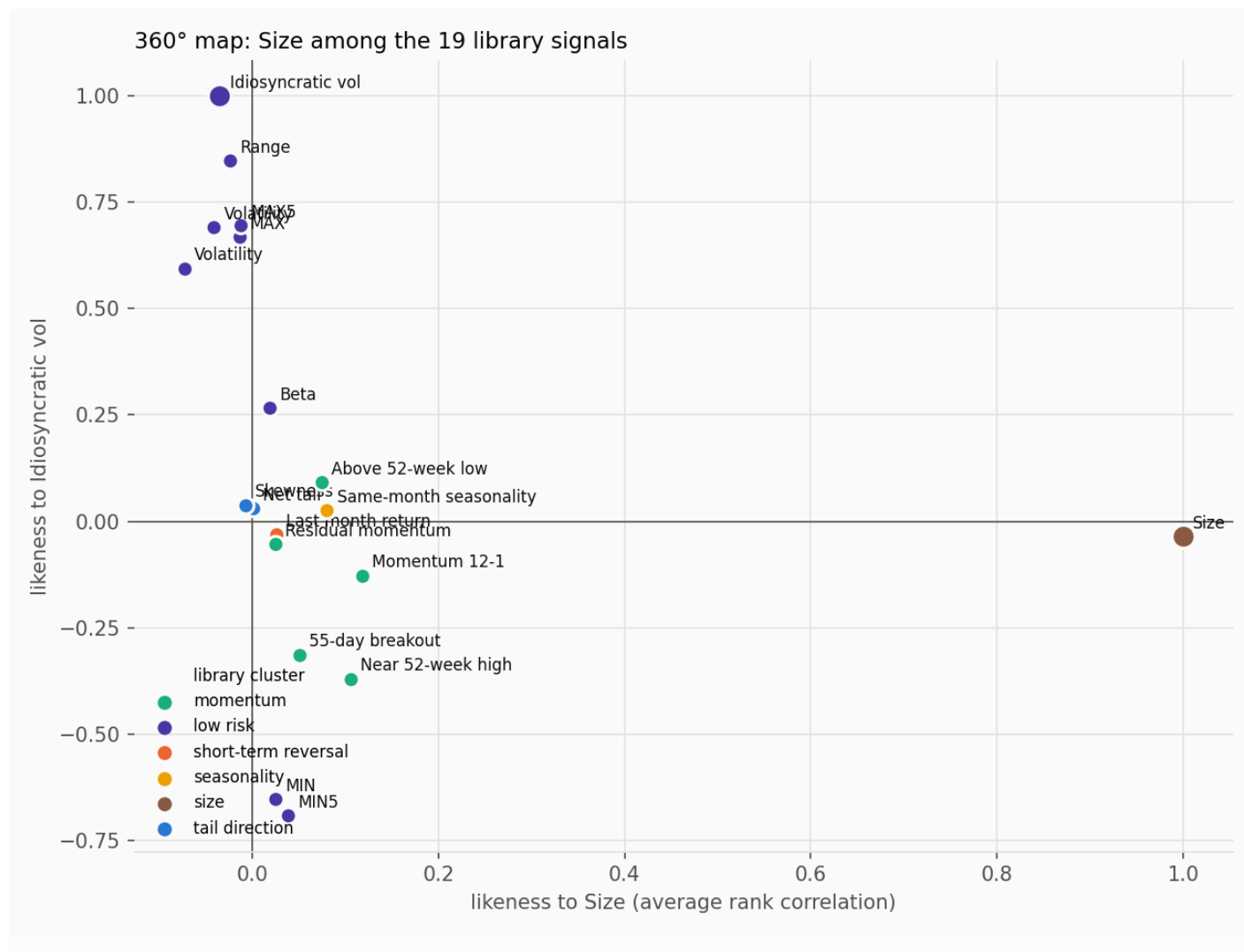


## D. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Idiosyncratic vol**.

| Signal (top minus bottom, raw sign) | Cluster  | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha t ≥ +2 / ≤ -2 |
|-------------------------------------|----------|------------------|-------------------|-------------------|------------------|-------------------------------|
| Size ( <b>this strategy</b> )       | size     | +1.00            | +1.00             | +0.1%             | -3.8% to +9.9%   | 1 / 0                         |
| Near 52-week high                   | momentum | +0.11            | +0.36             | +3.7%             | +9.3% to +22.3%  | 6 / 0                         |
| Momentum 12-1                       | momentum | +0.12            | +0.33             | +7.2%             | +4.3% to +19.4%  | 5 / 0                         |
| Volatility (252 days)               | low risk | -0.07            | -0.33             | +2.4%             | -12.9% to -4.8%  | 0 / 3                         |
| 55-day breakout                     | momentum | +0.05            | +0.32             | +1.4%             | +4.1% to +13.0%  | 4 / 0                         |
| Volatility                          | low risk | -0.04            | -0.32             | +1.1%             | -13.7% to -7.9%  | 0 / 5                         |
| Range (MAX-MIN)                     | low risk | -0.02            | -0.31             | -0.7%             | -15.4% to -4.8%  | 0 / 4                         |

| Signal (top minus bottom, raw sign) | Cluster             | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha $t \geq +2$ / $\leq -2$ |
|-------------------------------------|---------------------|------------------|-------------------|-------------------|------------------|-----------------------------------------|
| MIN5                                | low risk            | +0.04            | +0.30             | -0.9%             | +2.3% to +13.7%  | 4 / 0                                   |
| MIN (worst day)                     | low risk            | +0.02            | +0.30             | -0.7%             | +3.3% to +11.1%  | 4 / 0                                   |
| Idiosyncratic vol (mirror)          | low risk            | -0.04            | -0.29             | -3.6%             | -15.5% to -4.8%  | 0 / 4                                   |
| Beta                                | low risk            | +0.02            | -0.28             | +4.4%             | -12.6% to -4.2%  | 0 / 4                                   |
| MAX (best day)                      | low risk            | -0.01            | -0.25             | +0.1%             | -13.3% to -2.0%  | 0 / 2                                   |
| MAX5                                | low risk            | -0.01            | -0.24             | +0.6%             | -11.3% to -3.5%  | 0 / 2                                   |
| Last month return                   | short-term reversal | +0.03            | +0.22             | +1.8%             | -3.2% to +11.0%  | 2 / 0                                   |
| Above 52-week low                   | momentum            | +0.07            | +0.15             | +7.5%             | +5.2% to +13.6%  | 4 / 0                                   |
| Residual momentum                   | momentum            | +0.02            | +0.11             | +6.4%             | +8.0% to +12.2%  | 6 / 0                                   |
| Same-month seasonality              | seasonality         | +0.08            | +0.08             | -1.9%             | -9.4% to +6.6%   | 0 / 1                                   |
| Net tail (MAX+MIN)                  | tail direction      | +0.00            | +0.07             | -0.9%             | -5.2% to +5.2%   | 0 / 0                                   |
| Skewness                            | tail direction      | -0.01            | -0.06             | -0.4%             | -6.7% to +3.9%   | 0 / 2                                   |



**Spanning.** Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t     | R <sup>2</sup> | Neighbours with  t  ≥ 2                                         |
|----------|-----------------------------|-------|----------------|-----------------------------------------------------------------|
| US       | +3.3%                       | 1.27  | 0.72           | Momentum 12-1 +0.39 (+4.8)                                      |
| EU       | +1.4%                       | 0.40  | 0.17           | Idiosyncratic vol -0.30 (-2.2)                                  |
| UK       | -1.5%                       | -0.57 | 0.14           | Same-month seasonality -0.15 (-3.8), Momentum 12-1 -0.12 (-2.2) |
| DK       | +0.6%                       | 0.20  | 0.21           | Above 52-week low +0.22 (+2.2), Momentum 12-1 +0.24 (+2.3)      |
| SCANDI   | +0.7%                       | 0.21  | 0.14           | Volatility -0.27 (-2.0)                                         |
| World    | +1.6%                       | 0.80  | 0.54           | MIN5 -0.19 (-2.4)                                               |

3×3 sort, next-month return a year  
(US, EU, UK, SCANDI, World averaged)

|           |                       |        |                        |
|-----------|-----------------------|--------|------------------------|
| low Size  | 12.3%                 | 13.0%  | 11.6%                  |
| middle    | 11.1%                 | 12.5%  | 10.8%                  |
| high Size | 11.8%                 | 11.8%  | 9.6%                   |
|           | low Idiosyncratic vol | middle | high Idiosyncratic vol |

Holding Idiosyncratic vol fixed (down a column), moving from low to high Size changes the return by -0.5, -1.2 and -1.9 points a year; holding Size fixed (along a row), moving from low to high Idiosyncratic vol changes it by -0.7, -0.3 and -2.2. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -6% vs +11% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +0.1%, +0.3%, +0.2%, -0.9% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)

